

Cohand VR: Towards A Shareable Immersive Experience via Wearable Gesture Interface between VR Audiences and External Audiences

Yuetong Zhao
School of New Media Art and Design,
Beihang University, Beijing, China
15811316820@163.com

Shuo Yan
School of New Media Art and Design,
Beihang University, State Key
Laboratory of Virtual Reality
Technology and Systems, Beihang
University, Beijing, China
yanshuobit@gmail.com

Xukun Shen
School of New Media Art and Design,
Beihang University, State Key
Laboratory of Virtual Reality
Technology and Systems, Beihang
University, Beijing, China
xkshen@buaa.edu.cn

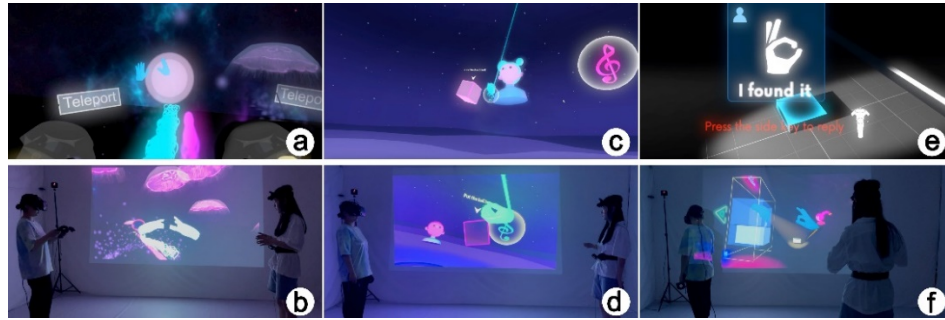


Figure 1: The three modes of Gesture-Based Interaction from the VR audience's perspective and the external audience's perspective, (a)(b) Spatial Localization, (c)(d) Object Interaction, (e)(f) Collaboration Interaction

ABSTRACT

Head-mounted displays (HMDs) increase immersion in virtual world, however which limits VR audiences' awareness of the external audiences that co-located in the same physical environment. In this paper, we superimposing the virtual world onto physical environment and propose a shareable VR experience between VR audiences and external audiences via a wearable interface, thereby enabling external audiences to observe and interact with the virtual objects by using hand gestures. Our system allows external audiences to explore the virtual scenario spatially and to generate asymmetric communication with VR audiences in three different modes. Research suggests that these design strategies could be adapted for future shareable VR experience.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • Interaction paradigms; • Virtual reality;

KEYWORDS

Sharing VR experience, Asymmetric interaction, Gesture-based interaction

ACM Reference Format:

Yuetong Zhao, Shuo Yan, and Xukun Shen. 2022. Cohand VR: Towards A Shareable Immersive Experience via Wearable Gesture Interface between VR Audiences and External Audiences. In *SIGGRAPH Asia 2022 Posters (SA '22 Posters)*, December 06–09, 2022, Daegu, Republic of Korea. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3550082.3564176>

1 INTRODUCTION

Head-mounted displays provide the user with a highly immersive VR experience. However, the HMD isolates the VR audience from the outside world, causing communication gaps with external audience who co-locate at the same physical environment. Researchers demonstrate novel systems that expand the virtual world into physical environment to augment a shared VR experience. This allows external audience to interaction with the VR audience [Kudo et al., 2021]. However, there are still some shortages. In some studies, the external audience's view is limited by the perspective of the VR audience's [Wang et al., 2020]. These affect the external audience's spatial perception of the virtual world. Furthermore, the external audience is usually considered as collaborators or competitors which are designed for asymmetric system [Gugenheimer et al., 2017]. That leads to a lower degree of freedom for the external audience to interact and explore in the virtual world.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '22 Posters, December 06–09, 2022, Daegu, Republic of Korea

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9462-8/22/12.

<https://doi.org/10.1145/3550082.3564176>

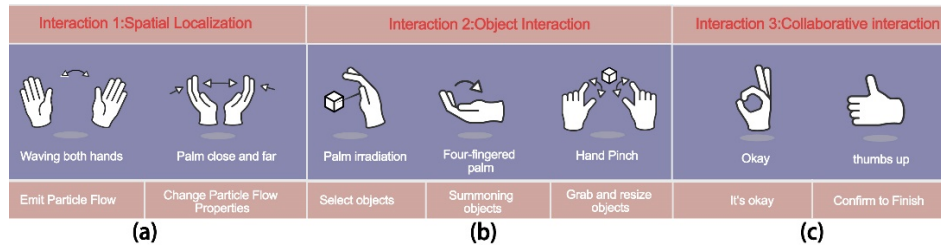


Figure 2: Gesture interaction instructions in three virtual scenarios

In this paper we propose a co-location, shareable immersive experience with three gesture design modes which include “Spatial Localization”, “Object Interaction” and “Collaboration Interaction” to enhance the external audience’s asymmetric interaction capability. The external audience uses a VIVE tracker that attached to his or her head to locate the relative position in the virtual scenario. An accompanying virtual camera projects the external audience’s perspective onto the physical environment. We design three modes of gesture interaction by using Leap Motion that tied around the external audience’s waist to interact with the VR audience in real-time.

2 THREE MODES OF GESTURE INTERACTION

2.1 Spatial Localization

In Scenario 1, we designed the interaction for both audiences to generate one-way particle streams from their own locations to the other to help perceive the spatial locations (Figure 1.a.b.). We designed two gestures: *waving hands* and *palms close and far* (Figure 2.a.). The external audience can generate a blue particle stream by waving hands to attract the attention of the VR audience. The size, color and speed of the particle stream will change accompanied by the distance between the external audience’s hands. When the VR audience notices, he or she can press the trigger button and change the distance between the VR handles to generate a purple particle stream as response. When both particle streams were triggered, two virtual avatars that representing both audiences will appear in the scene.

2.2 Object Interaction

In Scenario 2, we provided object interaction through generating interactive rays for both audiences (Figure 1.c.d.). We designed three hand gestures that help the external audience interact with virtual object: *select*, *summon*, and *grab* (Figure 2.b.). We set a task that asking both audiences of picking up specific color notes and throwing them to each other. The external audience can select the blue note through a generated blue ray by pinching with five fingers, then he or she can summon the note ball by flexing the four fingers and placed into the square to throw at the VR audience by pinching hand. The VR audience can press the handle’s trigger button to generate a purple ray and select the purple note. The VR audience can place the purple note into the square and pass it back to the external audience.

2.3 Collaborative Interaction

In Scenario 3, we designed an asymmetric collaborative game (Figure 1.e.f.). The external audience looks for the location of the puzzles and send gestures (Figure 2.c.) to inform the VR audience where the puzzles are. VR audience can teleport and pick up the puzzles in virtual scenario. Meanwhile, the external audience holds a searchlight in the left hand, the range and brightness of the searchlight can be changed by pinching the index finger and thumb from the right hand. The external audience can use the searchlight to find and illuminate the location of the puzzle and send an *Okay* gesture to inform the VR audience that the puzzle has been found. The VR audience could follow the light cues and use handles to teleport and pick up the puzzle, then press the trigger button as confirmation. The external audience would transfer the puzzle into the target position by presenting the *thumbs up* gesture.

3 CONCLUSIONS

In this paper, we design and implement a shareable immersive experience between the VR audience and external audience by using three modes of gesture-based interaction. A preliminary study shows that our system receives positive feedbacks on the sense of immersion, interesting and gesture interaction. It could help both audiences perceive each other’s interactions more clearly and give better feedback for collaboration. The participants provided additional suggestions for future implementation: 1) expand the physical space interaction scope of the external audience, 2) more natural gesture-based interaction, 3) participatory design for multiple audiences.

ACKNOWLEDGMENTS

This work is supported by the National Natural Science Foundation of China, under Grants 61802012.

REFERENCES

- Kudo, Y., Tang, A., Fujita, K., Endo, I., Takashima, K., & Kitamura, Y. 2021. Towards Balancing VR Immersion and Bystander Awareness. In *Proceedings of the ACM on Human-Computer Interaction*, 5(ISS), 1-22.
- Wang, C. H., Yong, S., Chen, H. Y., Ye, Y. S., & Chan, L. 2020. HMD light: Sharing In-VR experience via head-mounted projector for asymmetric interaction. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (pp. 472-486).
- Gugenheimer, J., Stemasov, E., Frommel, J., & Rukzio, E. 2017. ShareVR: Enabling co-located experiences for virtual reality between hmd and nonhmd users. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (pp. 4021-4033).